

EXPONENTIAL GAP-TIME ESTIMATION WITH CORRELATED RECURRENT EVENT MODELS: APPLICATION TO NEURAL FIRING DATA

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Summary: Time to occurrence of an event in a recurrent event data setting could be affected by many factors such as unobservable *frailty*, and *informative censoring*. Unobservable frailty induces a subject specific correlation among the inter-event times. Informative censoring controls the per subject accumulation of events. There are some analytical tools for estimating survivor function parameters in the presence of recurrent events. These generally lack the joint effect of correlation, informative censoring and randomness of the per subject observation window. Our method applies to correlated recurrent event data under informative censoring; specifically studies in which the survival time for a subject is censored because of deterioration of their physical condition or due to the accumulation of their events occurrences. In this paper, we approach the inter-event times estimation problem through a fully parametric baseline hazard model where recurrent

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events and censoring intensity are reconciled through the generalised Koziol-Green model for recurrent events. We further outline a general computational procedure for estimating parameters with correlated recurrent event data under informative censoring. The method is developed for exponential inter-event times. Algebraic formulas that extend it to Weibull inter-event times are proposed. We examine the sampling distribution properties of the estimators analytically and also *via* simulation. We apply our method to a neural activity biomedical data set.

1. Introduction

Areas of application of the methods developed herein are simplified through the following pictorial representation (Figure 1). In simple terms, a random sample of n units (experimental units or subjects) is monitored over time for a certain recurrent event. Units are censored while events are still expected to be observed. Censoring is informative and the number of events per unit (K_i , $i = 1, \dots, n$, say) is informative about the inter-event time distribution. One wishes to estimate the mean inter-event time amidst a potential within-unit events correlation and an informative monitoring aspect of the recurrent event.

Studies that involve recurrent event occur in many disciplines such as biomedical fields, public health, economics, engineering, and geology, just to name a few. In biomedical settings for instance, one could be interested in the recurrence of tumors, recurrence of migraine, epileptic seizures, or spikes arrival following a stimulus onset.

In public health, focus may be around the next epidemic of a specific type of influenza such as H1N1 or other variants; the next wave of some viral or infectious type of disease that cannot be eradicated.

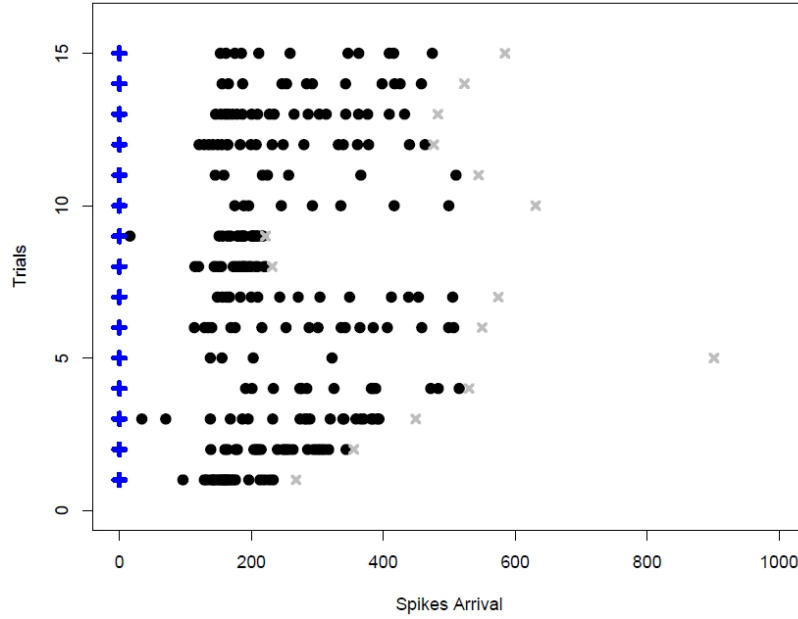


Figure 1. Example of recurrent event data (•), on 15 random cell lines (+), censoring represented by (x). In this neural activity study, the treatment is light stimulus and the response is the observed spike arrival times. Spikes are responses transmitted to the optic nerve.

In this paper we focus on the estimation of neural activity. To describe the experiment, 15 random cells each of which comes from a gridded area of a mouse retina were given a stimulus treatment with the goal to assess the stimulus effect on response transmission to the optic nerve. The observed response to stimulus is the signals/spikes arrival times obtained through a computerised system. Cells are censored after a certain duration τ_i , which varies from cell to cell through stimulus withdrawal. Our goal is to estimate the mean inter-event time. The direct implication is that, knowledge about the inter-event times may lead to better understanding of the neuron's language and

the complexity of neural activities. The setting lends itself to events recurring during a monitoring period.

Models for analysing these types of data are described in Hougaard (2000), Therneau and Hamilton (1997) and Therneau and Grambsch (2000). The estimators developed by ignoring correlation (in measurements within the same unit) is expected to yield a biased estimator of the marginal inter-occurrence time distribution if they were correlated (see for example Peña et al., 2001). Ignoring informative censoring leads to a loss in statistical efficiency (Vaart and Gill, 1993). Our goal is to develop a probabilistic model for analysing events recurrence behaviour and their inherent correlation. In our model, we incorporate various aspects pertaining to data accrual intensity, random variation window per unit, number of event occurrences per unit, the within unit events correlation and the censoring mechanism into one analytical tool.

To the best of our knowledge, the joint impact of correlation and informative censoring in recurrent event data has received less treatment in the literature. We use a parametric approach and propose to incorporate informative censoring through the generalised Kozil-Green (KG) model and to model correlation among inter-event times through a random frailty inherent to each unit (Koizol and Green, 1976). One advantage of the KG model is its ability to reconcile failure and censoring intensity through a non-complicated proportionality. The use of frailty can be traced back to Vaupel et al. (1979) and Hougaard (1984). More discussion on the use of frailty in the single and recurrent event settings can be found in Borgan (1984), Nielsen et al. (1992), Peña et al. (2001, 2007), Stocker and Peña (2007).

2. Random entities and model parameters

First, let us define the random entities in terms of the neural activity data.

These entities will be repeated throughout the paper.

- Unit i : i^{th} random grid around the retina, $i = 1, \dots, n$
- $N_i^\dagger(s)$: counts the number of messages sent to the optic nerve for grid i by calendar time s
- $Y_i^\dagger(s)$: indicates whether grid i is still sending messages to the optic nerve at time s
- $N_i^\tau(s)$: indicates whether grid i has already been censored
- τ_i : describes the observation window for grid i
- $R_i(w)$: measures time since last message was sent to the optic nerve
- K_i : counts the number of messages sent over $[0, \tau_i]$ for grid i .

2.1 Correlated recurrent event data and random entities

Consider n experimental units with the i^{th} unit monitored for the occurrence of a recurrent event over a period $[0, \tau_i]$, where the τ_i s are right-censoring random variables, i.i.d. with distribution function $G(t) = P(\tau_i \leq t)$ and survivor function $\bar{G} \equiv 1 - G$. For each unit i , we observe events at calendar times $0 = S_{i0} < S_{i1} < \dots$ with $K_i = \max\{k \in \mathbb{N} : S_{ik} \leq \tau_i\}$ the total number of event occurrences per unit. We denote by $T_{ij} \equiv S_{ij} - S_{i,j-1}$ the successive inter-event times. With recurrent events, the number of observed event occurrences is informative about the inter-event time distribution. Furthermore, the random variable $\tau_i - S_{iK_i}$ is the right-censoring random variable for the inter-event time T_{iK_i+1} for each unit and depends on previous inter-event times through S_{iK_i} . Informativeness is

implicit through the constraint

$$\sum_{j=1}^{K_i} T_{ij} = S_{iK_i} \leq \tau_i < \sum_{j=1}^{K_i+1} T_{ij} = S_{iK_i} + T_{iK_i+1}.$$

To describe correlation, we assume that there exists a per unit random variable Z_i , i.i.d. with distribution function $H(\cdot|\eta)$ where $\eta \in \mathbb{R}^p$.

Conditionally on Z_i , the T_{ij} s are assumed to be i.i.d. with distribution function F that has a conditional hazard rate and conditional survivor function given by

$$\lambda_F(t|Z_i) = Z_i \lambda_0(t); \quad \text{and} \quad \bar{F}(t|Z = z) = \{\bar{F}_0(t)\}^z = \{\exp(-\Lambda_0(t))\}^z,$$

respectively, where $\lambda_0(t)$ is the baseline hazard rate associated with F_0 .

As a consequence, the observable random vector for subject i is

$$(K_i, \tau_i, T_{i1}, \dots, T_{iK_i}, \tau_i - S_{iK_i}). \quad (1)$$

Since the Z_i s are unobserved, they are not included in the vector of observed entity (1). The goal of the analysis in this paper is to estimate the baseline survivor function and consequently the survivor function parameters under the assumption of informative monitoring based on the data in (1) and also under the assumption that there exists a unit-specific frailty.

2.2 The generalised Koziol-Green model for recurrent event

The generalised KG model for this recurrent event settings postulates that there exists a $\beta > 0$ such that $\bar{G}(t) = \bar{F}(t)^\beta$. In recurrent event setting, the parameter β determines the length of the monitoring period relative to the inter-event times and controls the events intensities during the monitoring period. For the exponential inter-event times model for instance, $\frac{1}{\beta}$ is the expected number of event occurrences during the monitoring period. The case $\beta \in (0, 1]$ is of more practical relevance since it leads to more observed recurrences, although $\beta > 1$ is also an admissible value. The smaller the

value of β , the higher the intensity of event occurrences during the monitoring period.

2.3 Relevant stochastic processes

For generality, we adopt the approach of counting processes and martingales (cf. Fleming & Harrington, 1991 and Andersen et al., 1993). Let $(\Omega, \mathcal{F}, P)$ be a complete probability space on which random entities are defined. For $i = 1, 2, \dots, n$; $j = 1, 2, \dots$; and $s \in \mathbb{R}$, we define the counting processes

$$\begin{aligned} N_i^\dagger(s) &= \sum_{j=1}^{\infty} I\{S_{ij} \leq s \wedge \tau_i\}; \quad Y_i^\dagger(s) = I\{\tau_i \geq s\} \\ \text{and } N_i^T(s) &= I\{\tau_i \leq s\}. \end{aligned} \quad (2)$$

We augment the probability space $(\Omega, \mathcal{F}, P)$ by the filtration $\mathbf{F} = \{\mathcal{F}_s : s \geq 0\}$ where $\mathcal{F}_s$ is the history of the process up to time s which also includes information at time 0. The filtration $\mathbf{F}$ is such that the processes in (2) are adapted and that $Y_i^\dagger$ is $\mathbf{F}$ -predictable. Also for each i , define the time since last event occurrence by $R_i(w) = w - S_{iN_i^\dagger(w-)}$. This is an $\mathbf{F}$ -adapted process, left-continuous, hence predictable. In general, we assume that $F_0 \in \mathcal{C} = \{F_0(\cdot, \theta) : \theta \in \Theta \subset \mathbb{R}^p\}$. However, in this paper we limit ourselves to the exponential distribution. Extension to the Weibull distribution is given in section 7. To model informative censoring, we use the generalised KG model which assumes the existence of $\beta > 0$ such that $\overline{G}(\cdot; \theta) = \overline{F}(\cdot; \theta)^\beta$.

From stochastic integration theory, the conditional $\mathbf{F}$ -compensator of $N_i^\dagger(s)$ and $N_i^T(s)$ are respectively given by

$$\begin{aligned}
A_i^\dagger(s; \boldsymbol{\theta} | Z_i) &= \int_0^s Y_i^\dagger(w) Z_i \lambda_0(w, \boldsymbol{\theta}) dw \quad \text{and} \\
A_i^\tau(s; \boldsymbol{\theta}, \beta | Z_i) &= \int_0^s Y_i^\dagger(w) Z_i \beta \lambda_0(w, \boldsymbol{\theta}) dw.
\end{aligned} \tag{3}$$

3. Estimation of the survivor function parameters

A common and convenient choice for the frailty distribution is a gamma distribution with shape and scale parameter α . This is needed in order to achieve model identifiability. With this choice for the distribution of the frailty, the common survivor function becomes

$$\bar{F}(t) = \left[\frac{\alpha}{\alpha + \Lambda_0(t)} \right]^\alpha. \tag{4}$$

The parameter α controls the degree of association among the inter-event times within each unit. For instance, as α increases (decreases), association decreases (increases). Moreover, if $\alpha \rightarrow \infty$, Z_i will converge to one in probability and we recover the model with independent inter-occurrence times. Because censoring is informative, the complete conditional likelihood process over $[0, s]$ is $L(s, \beta | z_1, \dots, z_n) = L$ and given by

$$\begin{aligned}
L &= \prod_{i=1}^n \left[\boldsymbol{\pi}_{w=0}^s \left\{ z_i Y_i^\dagger(w) \lambda_0(R_i(w)) dw \right\}^{dN_i^\dagger(w)} \right. \\
&\quad \times \left. \left\{ 1 - z_i Y_i^\dagger(w) \lambda_0(R_i(w)) dw \right\}^{1-dN_i^\dagger(w)} \right] \\
&\quad \times \boldsymbol{\pi}_{w=0}^s \left\{ z_i \beta Y_i^\dagger(w) \lambda_0(w) dw \right\}^{dN_i^\tau(w)} \\
&\quad \times \left\{ 1 - z_i \beta Y_i^\dagger(w) \lambda_0(w) dw \right\}^{1-dN_i^\tau(w)},
\end{aligned} \tag{5}$$

where $\boldsymbol{\pi}_{w=0}^s$ evaluates the product integral over a finite partition (with length dw) of the interval $[0, s]$. These dw will be cancelled when likelihood ratios are formed.

Note:

On a short note, observe that

$$L = \prod_{i=1}^n L_{i,1} \times L_{i,2},$$

where

$$\begin{aligned} L_{i,2} &= \boldsymbol{\pi}_{w=0}^s \left\{ z_i \beta Y_i^\dagger(w) \lambda_0(w) dw \right\}^{dN_i^\tau(w)} \\ &\quad \times \left\{ 1 - z_i \beta Y_i^\dagger(w) \lambda_0(w) dw \right\}^{1-dN_i^\tau(w)}. \end{aligned}$$

$L_{i,2}$ represents the contribution of the informative censoring part to the likelihood. With $L_{i,2}$, we allow the monitoring window to be random for each unit.

More on the likelihood

Since the Z_i s are unobserved, the likelihood in (5) and $\Lambda_0(\cdot | z_1, \dots, z_n)$ cannot be computed.

Applying Bayes' rule by multiplying the likelihood L by the joint density of $Z_1, \dots, Z_n$ (each Z_i has density $\alpha^\alpha z_i^{\alpha-1} \exp\{-\alpha z_i\} / \Gamma(\alpha)$), we obtain the likelihood function

$$\begin{aligned} L(s, \beta; z_1, \dots, z_n) &= \prod_{i=1}^n \beta^{N_i^\tau(s)} \frac{\alpha^\alpha}{\Gamma(\alpha)} z_i^{N_i^\dagger(s) + N_i^\tau(s) + \alpha - 1} \\ &\quad \times \boldsymbol{\pi}_{w=0}^s \left\{ Y_i^\dagger(w) \lambda_0(R_i(w)) \right\}^{dN_i^\dagger(w)} \boldsymbol{\pi}_{w=0}^s \\ &\quad \cdot \left\{ \beta Y_i^\dagger(w) \lambda_0(w) \right\}^{dN_i^\tau(w)} \\ &\quad \times \exp \left\{ -z_i \left(\alpha + \int_0^s Y_i^\dagger(w) \{ \lambda_0(R_i(w)) \right. \right. \\ &\quad \left. \left. + \beta \lambda_0(w) \} dw \right) \right\}. \end{aligned}$$

This expression is proportional to that of a gamma distribution with shape parameter $N_i^\dagger(s) + N_i^\tau(s) + \alpha$ and scale parameter $\alpha +$

$\int_0^s Y_i^\dagger(w) \{\lambda_0(R_i(w)) + \beta \lambda_0(w)\} dw$. As a consequence, given α , β , and $\Lambda_0(\cdot)$, and the observed values of the processes $Y_i^\dagger(w)$, $N_i^\dagger(w)$, $N_i^\tau(w)$ for $w \in [0, s]$, we have

$$\begin{aligned} E \left\{ Z_i | \alpha, \beta, \Lambda_0(\cdot), \{Y_i^\dagger(w), N_i^\dagger(w), N_i^\tau(w); i = 1, \dots, n; w \in [0, s]\} \right\} \\ = \frac{N_i^\dagger(s) + N_i^\tau(s) + \alpha}{\alpha + \int_0^s Y_i^\dagger(w) \{\lambda_0(R_i(w)) + \beta \lambda_0(w)\} dw}. \end{aligned} \quad (6)$$

To obtain an estimate of the parameters α , $\Lambda_0(\cdot)$ and β , one can integrate out $z_1, \dots, z_n$ in $L(s, \beta; z_1, \dots, z_n)$ and the corresponding likelihood could be used to obtain the estimates of the model parameters. By doing so, we obtain the partial likelihood process

$$\begin{aligned} L(s) &= \prod_{i=1}^n \frac{\Gamma(N_i^\dagger(s) + N_i^\tau(s) + \alpha)}{\Gamma(\alpha)} \\ &\quad \times \left\{ \frac{\alpha}{\alpha + \int_0^s Y_i^\dagger(w) \{\lambda_0(R_i(w)) + \beta \lambda_0(w)\} dw} \right\}^{N_i^\dagger(s) + N_i^\tau(s) + \alpha} \\ &\quad \times \boldsymbol{\pi}_{w=0}^s \left\{ \frac{Y_i^\dagger(w) \lambda_0(R_i(w))}{\alpha} \right\}^{dN_i^\dagger(w)} \\ &\quad \times \boldsymbol{\pi}_{w=0}^s \left\{ \frac{Y_i^\dagger(w) \lambda_0(w)}{\alpha} \right\}^{dN_i^\tau(w)} \beta^{N_i^\tau(s)}. \end{aligned}$$

4. Estimation under the exponential distribution

Figure 2 displays the variation in cells' reaction to stimulus. It is a graph of time versus the accumulation of spikes. The graph attests to cell difference in reaction to stimulus, and also suggests that an exponential fit is reasonable.

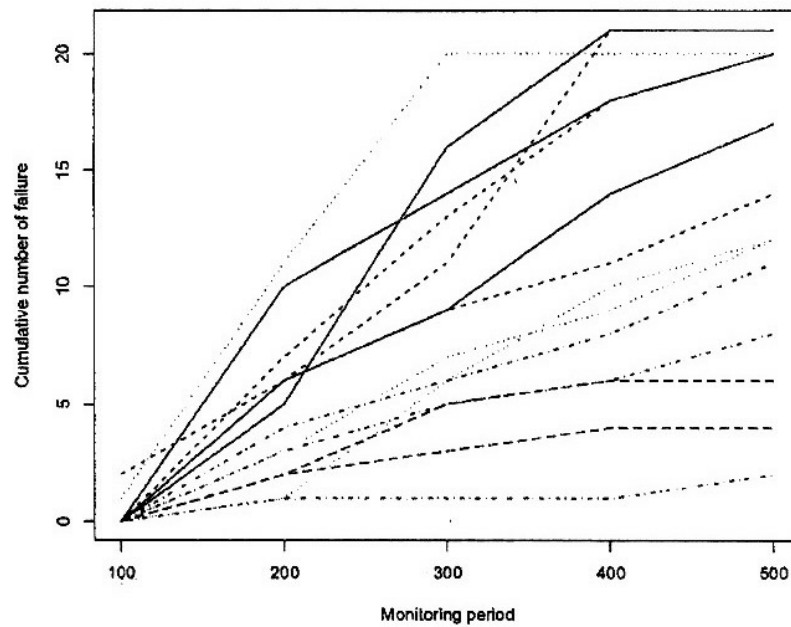


Figure 2. Neural activity data. Cumulative number of failures versus monitoring window.

With exponential inter-event times, the hazard rate is $\lambda_0(w) = \lambda_0(R_i(w)) = \theta$.

Based on previous derivations, the partial likelihood process is given by

$$\begin{aligned}
 L(s) &= \prod_{i=1}^n \frac{\Gamma(N_i^\dagger(s) + N_i^\tau(s) + \alpha)}{\Gamma(\alpha)} \\
 &\quad \times \left\{ \frac{\alpha}{\alpha + \theta(\beta + 1) \int_0^s Y_i^\dagger(w) dw} \right\}^{N_i^\dagger(s) + N_i^\tau(s) + \alpha} \\
 &\quad \times \left(\frac{\theta}{\alpha} \right)^{N_i^\dagger(s) + N_i^\tau(s)} \boldsymbol{\pi}_{w=0}^s \left[Y_i^\dagger(w) \right]^{dN_i^\dagger(w) + dN_i^\tau(w)} \\
 &\quad \times \beta^{N_i^\tau(s)};
 \end{aligned} \tag{7}$$

and

$$\begin{aligned}
 &E \left\{ Z_i | \alpha, \beta, \Lambda_0(\cdot), \{Y_i^\dagger(w), N_i^\dagger(w), N_i^\tau(w); i = 1, \dots, n; w \in [0, s]\} \right\} \\
 &= \frac{K_i + 1 + \alpha}{\alpha + (1 + \beta)\theta\tau_i}.
 \end{aligned} \tag{8}$$

From (4), the estimate of the common marginal survivor function $\hat{\bar{F}}(t)$ is given by

$$\hat{\bar{F}}(t) = \left[\frac{\hat{\alpha}}{\hat{\alpha} + \hat{\theta}t} \right]^{\hat{\alpha}}, \tag{9}$$

where the estimate of α and θ are obtained through an adaptation of the EM algorithm. Define $N_i^\dagger(s)$, $N_i^\tau(s)$, and $Y_i^\dagger(s)$ as in (2). Also, recall the conditional expectation of the frailties as stated in (6). Write $N_i = N_i^\dagger + N_i^\tau$ to condense the formula; then, from (7)

$$\begin{aligned}
 \log(L(s)) &= \sum_{i=1}^n \left\{ \log(\Gamma[N_i(s) + \alpha]) - \log(\Gamma[\alpha]) + [N_i(s) + \alpha] \right. \\
 &\quad \left. \log \left(\frac{\alpha}{\alpha + \theta(\beta + 1)(s \wedge \tau_i)} \right) \right\} \\
 &\quad + \sum_{i=1}^n N_i(s) \log \left(\frac{\theta}{\alpha} \right).
 \end{aligned}$$

There is no analytical solution for the value $\hat{\alpha}$ for which the likelihood function is maximised. An approximate value $\hat{\alpha}$ may be obtained through Newton-Raphson iteration. Defining Ψ as the di-gamma function and $\dot{\Psi}$ as the derivative of the di-gamma function with respect to α , it can be shown that the first and the second derivatives of the log likelihood as a function of α can be easily written as :

$$\begin{aligned} \frac{\partial \log(L(s))}{\partial \alpha} &= \dot{l}(\alpha, s) = \sum_{i=1}^n \left\{ \Psi[N_i(s) + \alpha] - \Psi[\alpha] \right. \\ &\quad \left. + \log \left(\frac{\alpha}{\alpha + \theta(\beta + 1)(s \wedge \tau_i)} \right) \right\} \\ &\quad + \sum_{i=1}^n \left\{ \frac{\theta(\beta + 1)(s \wedge \tau_i) - N_i(s)}{\alpha + \theta(\beta + 1)(s \wedge \tau_i)} \right\}. \\ \frac{\partial^2 \log(L(s))}{\partial \alpha^2} &= \ddot{l}(\alpha, s) = \sum_{i=1}^n \left\{ \dot{\Psi}[N_i(s) + \alpha] - \dot{\Psi}[\alpha] \right. \\ &\quad \left. + \frac{N_i(s)\alpha + [\theta(\beta + 1)(s \wedge \tau_i)]^2}{\alpha[\alpha + \theta(\beta + 1)(s \wedge \tau_i)]^2} \right\}. \end{aligned}$$

The Newton-Raphson algorithm with starting value $\hat{\alpha}_0$ estimates α_m through the iterative process

$$\alpha_m = \alpha_{m-1} - \frac{\dot{l}(\alpha_{m-1})}{\ddot{l}(\alpha_{m-1})},$$

and yields convergence when $|\alpha_m - \alpha_{m-1}| < \epsilon$; where ϵ is the tolerance.

The α value obtained from this process is used in the following EM adaptation.

Algorithm

1. Find a naive estimate of the hazard $\hat{\theta}_0$, say. The Expectation step of the algorithm proceeds as follow:

Given $\hat{\alpha}_0$ and $\hat{\theta}_0$, obtain estimates of the unobserved frailties as

$$\hat{Z}_i = \frac{N_i^\dagger(s) + N_i^\tau(s) + \hat{\alpha}_0}{\hat{\alpha}_0 + (1 + \beta)\hat{\theta}_0(s \wedge \tau_i)}; \quad i = 1, 2, \dots, n.$$

2. The M step finds an update $\hat{\theta}_1$ that maximises

$$\begin{aligned} L(s, \beta, \theta; z_1, \dots, z_n) &= \prod_{i=1}^n \beta^{N_i^\tau(s)} \frac{\alpha^a}{\Gamma(\alpha)} z_i^{N_i^\dagger(s) + N_i^\tau(s) + \alpha - 1} \\ &\quad \times \boldsymbol{\pi}_{w=0}^s \left\{ Y_i^\dagger(w) \theta \right\}^{d_{N_i^\dagger(w)}} \\ &\quad \times \boldsymbol{\pi}_{w=0}^s \left\{ \beta Y_i^\dagger(w) \theta \right\}^{d_{N_i^\tau(w)}} \\ &\quad \times \exp \left\{ -z_i \left(\alpha + \int_0^s Y_i^\dagger(w) \{ \theta + \beta \theta \} dw \right) \right\}; \end{aligned}$$

namely

$$\hat{\theta} = \frac{\sum_{i=1}^n N_i(s)}{\sum_{i=1}^n \hat{z}_i (\beta + 1) (s \wedge \tau_i)}. \quad (15)$$

3. Alternating the E and M aspects of the algorithm, the steps come as follow:

- Find a starting value for θ say θ_0 .
- Use the starting value of θ to estimate the mle for α through Newton-Raphson; say α_0 .
- Use the starting value of α_0 and θ_0 to estimate the expectation of the Z_i s.
- Use this update to estimate $\hat{\alpha}_1$ that maximises the likelihood $l(s, \alpha)$; then cycle back to estimate the new frailties $\hat{Z}_i$ s, then estimate $\hat{\theta}_{(\cdot)}$, then $\hat{\alpha}_{(\cdot)}$ until the estimates stabilise.

$\hat{\theta}_{(\cdot)}$ and $\hat{\alpha}_{(\cdot)}$ are respectively the estimated values of θ and α at iteration $(\cdot)$. We used both the conditional and the marginal likelihoods, i.e. $L(s | z_1, \dots, z_n)$ and $L(s)$ respectively, in our estimation. The marginal likelihood function yields a first-hand $\hat{\alpha}$ value through Newton Raphson or Simplex method of Nedler and Mead (1965). Note that this value has been

updated in subsequent iterations. In the E-step, $\hat{Z}$ is easily obtainable from a starting value $\hat{\theta}_0$ because the conditional likelihood is linear in z . Down to the M-step itself, we take a double step. Given the estimated $\hat{Z}_i$, we obtain an updated value of $\hat{\theta}_{(\cdot)}$, then another updated estimate $\hat{\alpha}$ of α is obtained through maximisation of the profile marginal log likelihood. Thus, our estimation procedure is an EM adaptation as the ones described by Nielsen et al. (1992); Peña et al. (2001). For other relevant estimation approaches of model parameters in the presence of frailty, see Manda (2001) which describes an estimation under both frequentist and Bayesian models, and Sastry (1997) in which EM algorithms were applied to independent components of a profile log likelihood.

5. Simulation results

We conduct a simulation to assess the accuracy of our model estimates. The model was implemented using the R software on LINUX platform. We operate under the following settings: The monitoring parameter $\beta \in (0.3; 0.5; 0.7)$. Sample sizes of 50, 100, 500, and 1 000 were considered in order to explore small sample and asymptotic bias of the estimates. The frailty model used is Gamma(α, α); for simplicity, ease of interpretability, and model identifiability (Nielsen et al., 1992). We choose α ranging from 0.5 to 10; while θ varies from 1 through 10. For each of these combinations, we run 1 000 replications to assess the validity of our model. The calendar times considered were $s = 5; 10; \text{ and } 20$. Figures 3 and 4 give us pictorial representations of how our model performs relative to true estimates. Because the results are similar across simulation, we restrict ourselves to showing one set in small sample ($n = 50$) and another set in large sample ($n = 500$). One panel estimates α values for a fixed value of $\theta = 4$, the other panel estimates the fixed value of

θ when α varies. On the right panel of each figure, we add an horizontal line indicating the actual value of θ used for this specific simulation. Goodness-of-fit for the estimates of α was assessed by regressing the estimated values on the actual; and variability assessed by the residual standard error. Since there are a few outliers in the estimates of α , we fit this regression on the log scale. The residual standard error is 0.38 and $r^2 = .8937$. The simulation results for α with θ fixed can be summarised as follows:

- For α , estimation is more accurate when n is large.
- Regardless of n , small values of α are estimated with the smallest variance.
- Variability around the estimated $\hat{\alpha}$ increases with actual α .

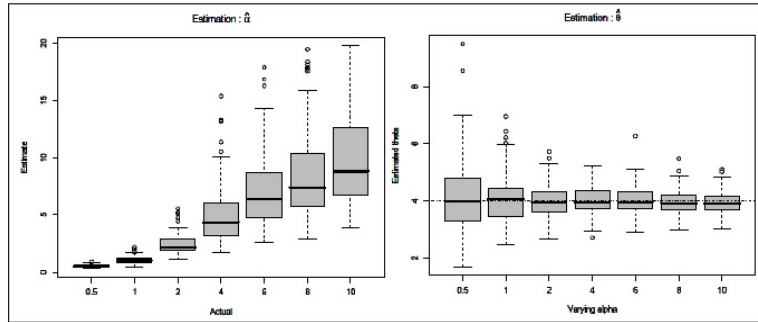


Figure 3. Performance study with $s = 5, \theta = 4, \beta = .3, \alpha \in (.5, \dots, 10)$, sample size $n = 50$.

1. The left panel estimates α 2. The right panel estimates θ when α varies.

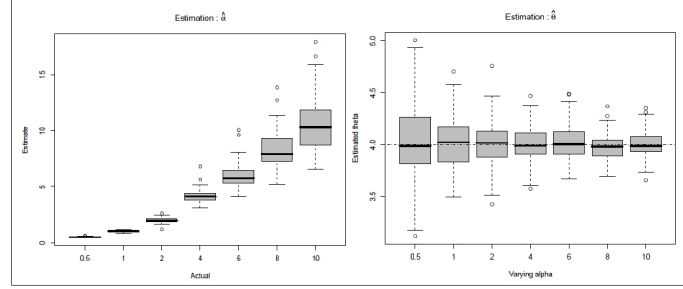


Figure 4. Performance study with $s = 5$, $\theta = 4$, $\beta = .3$, $\alpha \in (.5, \dots, 10)$, sample size $n = 500$. The left panel estimates α . The right panel estimates θ when α varies.

The implication of the second bullet point is that when intraclass correlation is high, our model is more likely to signal this; and doing so accurately. This property makes our approach more suited to model the correlated recurrent events it has been designed for. In addition, the estimate of θ is unbiased and remains so when α varies.

In another simulation setting, we consider values of $\theta \in (0, 50]$ and vary α at random on the positive line. Since large values of α are admissible but not practical (tend to the independent case with probability 1), we bound α to 50 for practical relevance. We then estimate $\hat{\theta}$ from our model and assess its goodness-of-fit. Figure 5 displays the estimates $\hat{\theta}$ for monitoring parameter $\beta = 0.3$. Similar results are obtained across simulations. The estimates for θ are precise with less variance. Table 1 reports r^2 as measure of goodness-of-fit and residual standard error (*r.s.e*) from regressing estimates on actual values of θ for a few selected α values. On a side note, for the case $\beta = .3$, the estimate $\hat{\beta} = 0.3004$ and the standard error is 0.002.

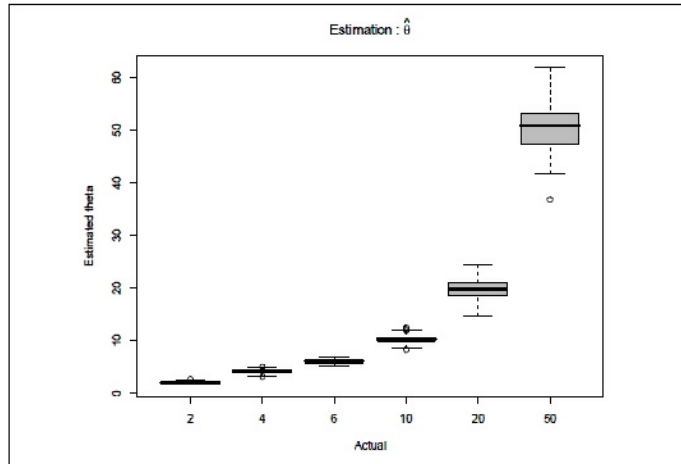


Figure 5. Boxplots from 1000 replications estimating θ . Data generated according to true values of $\theta \in (2, 4, 6, 10, 20, 50)$; $\beta = .3$; and $n = 500$. True parameter on x -axis, estimated parameter on y -axis.

Table 1 Goodness-of-fit table with r^2 and residual standard error for actual versus estimated θ

Simulation Results $s = 5$; $n = 500$; $\beta = 0.3$							
$\alpha =$	0.5	1	2	4	6	8	10
r^2	0.9933	0.9964	0.9977	0.9986	0.9988	0.9990	0.9990
$r.s.e$	0.0668	0.0631	0.0512	0.0398	0.0363	0.0337	0.0330

5.1 Informative vs. non-informative censoring

Many authors have warned against the danger of ignoring informative censoring in survival parameter estimation (see Vaart & Gill, 1993; Adekpedjou et al., 2010). In order to assess the effect of informative censoring on estimation, we conduct a simulation to report the bias associated to each method (estimation with informative censoring and without it), along with the

relative efficiency. The simulation setting consists of varying a true θ in the interval $[1, 50]$ and estimating $\hat{\theta}$ via these two methods – $\hat{\theta}_i$ for informative censoring and $\hat{\theta}_n$ without. Next, each simulation setting is coupled with a varying association parameter α to identify the bias distribution as association weakens. The monitoring parameter considered is $\beta = 0.3$ and the number of replication for the bias distribution is 100 each. The results are presented in the following series of boxplots (Figure 6). The grey boxes represent bias estimation under informative censoring while the white boxes ignore it.

Note that, for the most part, the parameter θ has been severely underestimated when ignoring information censoring. In addition, the relative efficiency shows that the estimation procedure under informative censoring always dominates (see Table 2); and that statistical efficiency is lost through this ignorance.

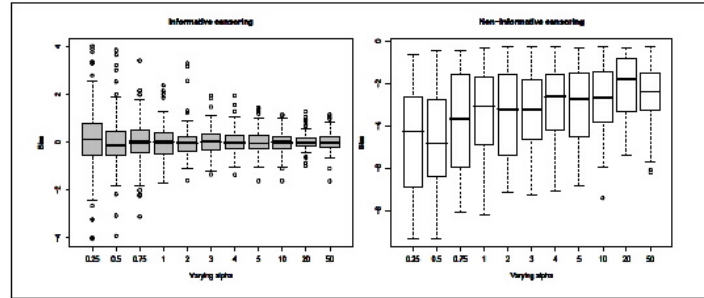


Figure 6. Bias assessment study $s = 5$, true $\theta \in 1, \dots, 50$, $\beta = .3$, $\alpha \in (.25, \dots, 50)$, sample size $n = 100$.
 1. Left panel: estimation under informative censoring
 2. Right panel: estimation under non informative censoring.

Table 2 Ratio of estimation variances

Setting: True θ randomly $\epsilon [1, 50]$, $\beta = 0.3$, $s = 5$, $n = 100$											
α	0.25	0.75	0.5	1	2	3	4	5	10	20	50
$var(\hat{\theta}_n) / var(\hat{\theta}_i)$	2.50	3.65	5.73	7.84	8.00	8.23	9.77	10.14	10.91	16.63	10.58

6. Application: Neural activity

The data on neural activity were generously supplied by the ophthalmology and neurology research laboratory at the University of Iowa Hospital and Clinics. It is a murine study where 15 random cells in gridded areas of the retina were given a stimulus treatment, with the goal to assess the stimulus effect on response transmission to the optic nerve. The observed response to stimulus is the signals/spikes arrival times obtained through a computerised system. Censoring occurs at a random duration τ_i . Our goal is to estimate the mean inter-event time. The direct implication is that knowledge about the mean inter-event time may lead to better understanding the neuron's language and the complexity of neural activities. The p -value for assessing between cells variation is smaller than 0.001; suggesting heterogeneity in cells' response to stimulus. Our method is applied to the data to estimate the inter-event times distribution based on an exponential probability model with parameter θ , in which the correlation between recurrent events is modelled by a frailty parameter α and the censoring mechanism modelled through the generalised KG model. Recall that the estimates of θ and the monitoring parameter β are:

$$(\hat{\theta}, \hat{\beta})(s) = \left(\frac{\overline{N^\dagger}(s)}{\overline{z(\tau \wedge s)}}, \frac{\overline{N^\tau}(s)}{\overline{N^\dagger}(s)} \right).$$

Note that $\hat{\beta}$ can be obtained directly from the sample at calendar time s through knowledge about all units under study and the number of recurrences up till time s . To estimate θ from the sample, one needs knowledge about the frailties. Since these are unobserved, we need a naive estimate of θ , $\hat{\theta}_0$

say, to carry into the EM-type of estimation. Note that the true estimate will be recuperated at convergence of the EM algorithm. Recall that $\overline{N^\dagger}(s) = \frac{1}{n} \sum_{i=1}^n N_i^\dagger(s)$; $\overline{N^\tau}(s) = \frac{1}{n} \sum_{i=1}^n N_i^\tau(s)$ and that $\overline{z(\tau \wedge s)} = \frac{1}{n} \sum_{i=1}^n z_i(\tau_i \wedge s) = \frac{1}{n} \sum_{i=1}^n z_i \int_0^s Y_i^\dagger(w) dw$. We set the starting value $\hat{\theta}_0$ to $\frac{\overline{N^\dagger}(s)}{\overline{z(\tau \wedge s)}}$; which illustrates the scenario where subjects have the same frailty (this assumption is less likely to occur, but is good enough to yield starting values for convergence). With these two estimates, we use the Newton-Raphson method to find $\hat{\alpha}$. With knowledge about $\hat{\alpha}$, the frailties $\hat{z}_i$ are estimated; then we cycle back to estimating $\hat{\theta}$ and $\hat{\alpha}$, then $\hat{z}_i$, until these estimates stabilise. Table 3 gives the estimates of the parameters; two of which (α, θ) are used for the survivor function; and the other (β), reciprocal of which approximates the average number of events during the monitoring period. The standard errors were obtained by bootstrapping. With knowledge about the parameters, one can build the survivor function and estimate the mean inter-event time by finding the integral of the survivor function over the domain of neural activity (needless to remind the reader that for every positive random variable T , with survivor function S , $E(T) = \int S(t) dt$). The mean inter-event time, over the domain of neural activity, is estimated to 32.4605 milliseconds.

Table 3 Parameter estimates for the murine neural activity data

Parameters	θ	α	β
Estimates	0.0489	2.55	0.05
Standard error	0.0401	1.06	0.0109

The small value of $\hat{\alpha}$ suggests a within cell correlation of spike arrivals over the monitoring period. It further attests that a high percentage of the total time variability is explained by cell differences in response to stimuli. The parameter estimate $\hat{\beta}$ tells us that on average, each gridded area has sent roughly 20 messages to the optic nerve during the monitoring period. The

combination of $\hat{\alpha}$ and $\hat{\theta}$ have contributed to estimating the overall average delay between these messages. The delay time is essential to estimating the neural code transmission to the optic nerve, and further exploring the complexities of the neural language.

7. Extension and comparison

7.1 Extension to Weibull

The Weibull inter-event times allow the hazard to depend on t . We outline below the algebraic properties of Weibull parameters and the steps to follow to estimate them. In the case of Weibull parameters, three main algebraic formulas drive the estimation procedure. Knowing that Weibull distribution has two parameters (shape θ_1 and scale θ_2), defining $\boldsymbol{\theta} = (\theta_1, \theta_2)$, the cumulative hazard rate through the KG model with monitoring parameter β is given by

$$\Lambda_G(t; \boldsymbol{\theta}, \beta) = Z\beta\lambda_0(t, \boldsymbol{\theta}),$$

where $\lambda_0(t, \boldsymbol{\theta}) = \theta_1\theta_2(\theta_2 t)^{\theta_1-1}$ is the baseline hazard function. There is no closed-form expression for estimating θ_1 and θ_2 by maximum likelihood even in the case of i.i.d. inter-event times with informative censoring. One can show, *via* some algebra, that when inter-event times T_{ij} follow the Weibull distribution with shape parameter θ_1 and scale parameter θ_2 , under the frailty model, both Weibull parameters are linked by

$$\hat{\theta}_2 = \left[\frac{\sum_{i=1}^n K_i}{\sum_{i=1}^n z_i \sum_{j=1}^{K_i+1} T_{ij}^{\hat{\theta}_1}} \right]^{1/\hat{\theta}_1} = \left\{ \frac{K_{\bullet}}{W_{z_i}(\hat{\theta}_1)} \right\}^{1/\hat{\theta}_1};$$

where $K_{\bullet} = \sum_{i=1}^n K_i$; K_i is the number of events per subject, and $W_{z_i}(\theta_1) = \sum_{i=1}^n z_i \sum_{j=1}^{K_i+1} T_{ij}^{\theta_1}$.

One can also show that the maximum likelihood estimator for θ_1 satisfies the relationship

$$\begin{aligned} & (n + K_\bullet) + \sum_{i=1}^n \left\{ \log \tau_i^*(\theta_1) + \sum_{j=1}^{K_i} \log T_{ij}^*(\theta_1) \right\} \\ &= n \frac{\sum_{i=1}^n z_i \tau_i^*(\theta_1) \log \tau_i^*(\theta_1)}{\sum_{i=1}^n z_i \tau_i^*(\theta_1)} + \sum_{i=1}^n z_i \sum_{j=1}^{K_i+1} (T_{ij}^*(\theta_1) \log T_{ij}^*(\theta_1)); \end{aligned}$$

where $\tau_i^*(\theta_1) = K_\bullet \frac{\tau_i^{\theta_1}}{W(\theta_1)}$ and $T_{ij}^*(\theta_1) = K_\bullet \frac{T_{ij}^{\theta_1}}{W(\theta_1)}$; and that β estimates to

$$\hat{\beta} = \frac{\sum_{i=1}^n N_i^\tau(s)}{\sum_{i=1}^n \int_0^s Y_i^\dagger(w) z_i \lambda_0(w, \theta) dw}.$$

An estimation procedure parallel to the one used in this paper can be employed by finding a starting value for θ_1 , estimating the Z_i' s, finding the maximum likelihood estimate of θ_1 , then θ_2 and finally β . Thus, the survival function, in this case, will be estimated by

$$\hat{\hat{F}}(t) = \left[\frac{\hat{\alpha}}{\hat{\alpha} + (\hat{\theta}_2 t)^{\theta_1}} \right]^{\hat{\alpha}}, t \in \mathfrak{R}_+.$$

Note that in the Weibull case, the asymptotic variance around the parameter estimates may not be available analytically. The renewal function associated with the inter-event times F (i.e. the distribution of S_{in}), does not have a closed-form when the inter-event time distribution differs from the exponential or the gamma.

7.2 Comparative study and concluding remarks

In advancing a new methodology, one wishes to compare to an existing standard. The existing standard we choose as benchmark for comparative study is the i.i.d. exponential inter-event times estimator with informative

censoring as outlined in Adekpedjou et al. (2010). The mle estimator of the inter-event times, in i.i.d. case, is the occurrence-exposure rate $\hat{\theta}_{i.i.d} = \frac{\sum K_i}{\sum \tau_i} = \frac{\bar{K}}{\bar{\tau}}$. The mle estimate of the monitoring parameter is $\hat{\beta} = \frac{n}{\sum_{i=1}^n K_i} \doteq \frac{1}{\bar{K}}$. The limiting asymptotic variances are given by

$$\left[\frac{I(\beta, \theta)}{n} \right]^{-1} = \begin{pmatrix} \theta^2 \beta & -\theta \beta^2 \\ -\theta \beta^2 & \beta^2(\beta + 1) \end{pmatrix} \doteq \begin{pmatrix} \frac{\bar{K}+1}{\bar{K}^3} & -\frac{1}{\bar{K}} \\ -\frac{1}{\bar{K}} & \frac{\bar{K}}{\bar{\tau}^2} \end{pmatrix}.$$

In the presence of frailties, we estimate the monitoring parameter and the hazard to be

$$(\hat{\theta}, \hat{\beta})(s) = \left(\frac{\bar{N}^\dagger(s)}{z(\tau \wedge s)}, \frac{\bar{N}^\tau(s)}{\bar{N}^\dagger(s)} \right) \doteq \left(\frac{\bar{K}}{z\bar{\tau}}, \frac{1}{\bar{K}} \right) \quad (16)$$

with asymptotic variance, and exact variance (in s) given respectively by:

$$\left[\frac{I(\beta, \theta)_{(n,s)}}{n} \right]^{-1} \doteq \begin{pmatrix} \frac{[1+\bar{K}]}{\bar{K}^3} & \frac{-1}{\bar{K}[z\bar{\tau}]} \\ \frac{-1}{\bar{K}[z\bar{\tau}]} & \frac{\bar{K}}{[z\bar{\tau}]^2} \end{pmatrix} \quad (17)$$

$$\left[\frac{I(\beta, \theta)}{n} \right]^{-1} \doteq \begin{pmatrix} \frac{[\bar{N}^\dagger(s) + \bar{N}^\tau(s)] \bar{N}^\tau(s)}{[\bar{N}^\dagger(s)]^3} & \frac{-\bar{N}^\tau(s)}{\bar{N}^\dagger(s) z(\tau \wedge s)} \\ \frac{-\bar{N}^\tau(s)}{\bar{N}^\dagger(s) z(\tau \wedge s)} & \frac{\bar{N}^\dagger(s)}{[z(\tau \wedge s)]^2} \end{pmatrix} \quad (18)$$

(See technical appendix for detail).

No doubt, both estimators concur when frailty is 1; the independence case. When inter-event times are correlated under informative censoring, there is a loss of efficiency by neglecting this correlation. As an example, the relative bias of the parameter estimated under i.i.d. assumption ($\hat{\theta}_{1,0}$) relative to the correlated events parameter estimation ($\hat{\theta}_{1,1}$) is $\frac{\theta_{1,0}}{\theta_{1,1}} = \frac{\bar{z}\bar{\tau}}{\bar{\tau}} \neq 1$. The ratio of the variances, $\frac{\text{var}(\hat{\theta}_{1,0})}{\text{var}(\hat{\theta}_{1,1})} = \left[\frac{\bar{z}\bar{\tau}}{\bar{\tau}} \right]^2 \neq 1$. Observe that efficiency is preserved with respect to the monitoring parameter. Small values of frailties

tend to explode the variance of θ while high frailties decrease it. In addition, small duration recurrent event studies do not necessarily abide by asymptotic rules. In such settings, the variance around the monitoring parameter and the inter-event times may be estimated in an exact way through (18), and their efficiency with respect to the independent setting or the single event setting may be assessed. Formulas (16), (17), and (18) confirm Peña et al. (2001, 2007) that ignoring correlation could lead to bias.

Note also, that Manda and Meyer (2005), Schaubel and Cai (2005), Box-Steffensmeier and De Boef (2006), Huang and Liu (2007) for example, discuss the estimation of cumulative hazard and/or survivor function of inter-event times. However, these works do not take into account the effect of informative censoring and/or the dependent right-censoring random variable of the recurrent event data. Manda and Meyer (2005), for example, describes a time-dependent frailty model using recurrent failure time data with non-informative censoring. In their Bayesian approach, the model parameters are estimated using Markov chain Monte Carlo; and their technique applies to data observed in an interval $[0, \tau_i]$ for subject $i = 1, \dots, n$ where τ_i is predetermined fixed constant. Schaubel and Cai (2005) too have a fixed window $[0, \tau_i]$; Box-Steffensmeier and De Boef (2006) ignore the censoring tail assumptions; and Huang and Liu (2007) assume death and censoring mechanisms to be independent and noninformative about the gap times; which also differs from our approach in which randomness and informative censoring contribute to the construction of the likelihood function. This joint aspect of correlation and informative censoring makes our proposed approach more general. The presence of the renewal function in our estimator is natural since it represents the expected number of events on $[0, t]$ for fixed t . The last right-censored event is included in our model whereas it is not accounted for in the

current literature of correlated recurrent analysis except in the non-parametric approach of Peña et al. (2001).

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8. Technical appendix

The mle for θ is given by

$$\hat{\theta} = \frac{\sum_{i=1}^n (N_i^\dagger(s) + N_i^\tau(s))}{\sum_{i=1}^n \hat{z}_i(\beta + 1) \int_0^s Y_i^\dagger(w) dw}.$$

The log likelihood process in terms of the frailties z_i can be written as:

$$\begin{aligned} l(s; \theta, \beta) &= \sum_{i=1}^n \left\{ \sum_0^s \log[Y_i^\dagger(w) z_i \beta \theta] dN_i^\tau(w) \right. \\ &\quad \left. + \sum_0^s \log[Y_i^\dagger(w) z_i \theta] dN_i^\dagger(w) \right\} \\ &\quad - \sum_{i=1}^n \left\{ \int_0^s Y_i^\dagger(w) \beta z_i \theta dw + \int_0^s Y_i^\dagger(w) z_i \theta dw \right\}. \end{aligned}$$

$$\frac{\partial l(s; \theta, \beta)}{\partial \beta} = \frac{1}{\beta} \sum_{i=1}^n N_i^\tau(s) - \theta \sum_{i=1}^n z_i \int_0^s Y_i^\dagger(w) dw;$$

which yields

$$\hat{\beta}(\theta, s) = \frac{\sum_{i=1}^n N_i^\tau(s)}{\theta \sum_{i=1}^n z_i \int_0^s Y_i^\dagger(w) dw}.$$

Thus, after some algebra and substitution we obtain

$$(\hat{\beta}, \hat{\theta})(s) = \left(\frac{\overline{N^\tau}(s)}{\overline{N^\dagger}(s)}, \frac{\overline{N^\dagger}(s)}{z(\tau \wedge s)} \right)$$

which yields the observed Fisher information matrix

$$\frac{I(\beta, \theta)}{n} = \begin{pmatrix} \frac{\overline{N^\tau(s)}}{\beta^2} & \overline{z(\tau \wedge s)} \\ \overline{z(\tau \wedge s)} & \frac{\overline{N^\tau(s) + N^\dagger(s)}}{\theta^2} \end{pmatrix}.$$

Small n , asymptotic in s :

This fits the setting of small sample long-lasting recurrent events

$$\frac{I(\beta, \theta)}{n} \doteq \begin{pmatrix} \frac{1}{\beta^2} & \overline{z\tau} \\ \overline{z\tau} & \frac{1+\overline{K}}{\theta^2} \end{pmatrix}.$$

Small s , asymptotic in n :

$$\frac{I(\beta, \theta)}{n} \doteq \begin{pmatrix} \frac{[\overline{N^\dagger(s)}]^2}{\overline{N^\tau(s)}} & \overline{z(\tau \wedge s)} \\ \overline{z(\tau \wedge s)} & \frac{\overline{N^\tau(s) + N^\dagger(s)}}{[\overline{N^\dagger(s)}]^2} [\overline{z(\tau \wedge s)}]^2 \end{pmatrix}.$$

The determinant is $[\overline{z(\tau \wedge s)}]^2 \frac{\overline{N^\dagger(s)}}{\overline{N^\tau(s)}}$; which is always positive and different from zero as long as there is at least one event in the population under study.

So, the inverse Fisher information exists.

$$\left[\frac{I(\beta, \theta)}{n} \right]^{-1} \doteq \begin{pmatrix} \frac{[\overline{N^\dagger(s) + N^\tau(s)}] \overline{N^\tau(s)}}{[\overline{N^\dagger(s)}]^3} & \frac{-\overline{N^\tau(s)}}{\overline{N^\dagger(s)} \overline{z(\tau \wedge s)}} \\ \frac{-\overline{N^\tau(s)}}{\overline{N^\dagger(s)} \overline{z(\tau \wedge s)}} & \frac{\overline{N^\dagger(s)}}{[\overline{z(\tau \wedge s)}]^2} \end{pmatrix}.$$

Asymptotic in both n and s ; $n \rightarrow \infty$ and $s \rightarrow \infty$:

$$\frac{I(\beta, \theta)_{(n,s)}}{n} \doteq \begin{pmatrix} \overline{K}^2 & \overline{z\tau} \\ \overline{z\tau} & \frac{[1+\overline{K}]}{\overline{K}^2} [\overline{z\tau}]^2 \end{pmatrix}.$$

The determinant is $\overline{K} [\overline{z\tau}]^2$, which is different from zero as long as there is a recurrence; in which case,

$$\left[\frac{I(\beta, \theta)_{(n,s)}}{n} \right]^{-1} = \begin{pmatrix} \frac{[1+\overline{K}]}{\overline{K}^3} & \frac{-1}{\overline{K}[\overline{z\tau}]} \\ \frac{-1}{\overline{K}[\overline{z\tau}]} & \frac{\overline{K}}{[\overline{z\tau}]^2} \end{pmatrix}.$$

The variance of θ depends on the frailty while the variance of β needs only knowledge about $\overline{K}$.

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